

Ethics at the Frontier: Environmental Policy, Climate Engineering, and Sustainability

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Abstract

This article examines how climate engineering and environmental artificial intelligence (AI) can be governed as components of sustainable and equitable technological transformation. Through qualitative conceptual analysis of contemporary governance frameworks, it synthesizes four recurring dimensions—precaution, justice, participation, and accountability—and integrates them into an Integrated Precautionary Governance Model (IPGM). The framework addresses the governance challenges posed by uncertain, potentially irreversible, and unequally distributed environmental technologies. Applying the IPGM to environmental AI and climate engineering, the article argues that sustainability cannot be assessed solely through technical or environmental outcomes but must also encompass democratic legitimacy, distributive equity, and institutional accountability. The article concludes by identifying priorities for operationalising and empirically testing the framework.

Keywords

Climate engineering; environmental AI; environmental governance; sustainability; climate justice; precautionary governance

Introduction

Climate change presents what Stephen Gardiner (2011) has characterised as a confluence of dispersed causation, fragmented agency, and profound intergenerational inequity that resists resolution through conventional policy instruments alone. As emissions continue to rise despite decades of international commitment, governments and scientists have increasingly turned to two categories of technological response: climate engineering, the deliberate large-scale manipulation of the Earth's climate system, and the application of AI to environmental monitoring, modelling, and governance. Both domains hold transformative promise and carry significant risks that, if unaddressed, could entrench existing injustices, undermine democratic accountability, and generate new geopolitical instabilities. These characteristics make the governance of emerging environmental technologies a sustainability problem as much as a technological or ethical one: technologies intended to address environmental degradation may redistribute risks and benefits, alter relationships of political power, and influence the conditions under which sustainable development is pursued.

Recent developments have significantly expanded the ethical and governance debates surrounding these technologies. The Oxford Principles have been revisited and extended. The AGU's Ethical Framework Principles for Climate Intervention Research has established a broadly applicable ethical framework for climate-intervention research, while UNESCO's COMEST has issued its first comprehensive report on the ethics of climate engineering. In parallel, the integration of AI into environmental decision-making has generated a growing consensus that ethical accountability for AI systems cannot be separated from their environmental and social consequences (Ali et al., 2026; Zeleti, 2026). Yet the convergence of these developments also raises a broader sustainability question: under what institutional and

normative conditions can technological intervention contribute to sustainable and equitable societal transformation rather than reproduce existing environmental inequalities?

This article addresses the following research question: What governance principles are necessary to ensure that high-impact environmental technologies such as climate engineering and environmental AI contribute to sustainable and equitable societal transformation while limiting uncertain, irreversible, and unequally distributed harms?

The analysis proceeds from the observation that existing governance frameworks frequently address individual dimensions of this problem without integrating them into a common architecture. Climate-engineering governance has developed extensive attention to precaution, consent, public participation, and distributive justice, while environmental-AI governance has increasingly emphasised algorithmic accountability, bias mitigation, transparency, and sustainability assessment. These literatures overlap in their concern with uncertainty, unequal impacts, institutional responsibility, and the legitimacy of technological intervention, yet they are rarely examined together.

The article therefore makes two related contributions. First, it examines four governance dimensions that recur across these literatures—precaution, justice, participation, and accountability—and examines their relationship across climate engineering and environmental AI. Second, it integrates these dimensions into the Integrated Precautionary Governance Model (IPGM), a conceptual framework for evaluating whether high-impact environmental technologies are being governed in ways consistent with sustainability, democratic legitimacy, and distributive equity. The framework is intended as a conceptual contribution to sustainability governance and as a basis for future empirical research into the governance of emerging environmental technologies.

Materials and Methods

This article employs qualitative conceptual analysis to examine contemporary governance frameworks relating to climate engineering and environmental artificial intelligence (AI). Rather than reporting original empirical findings, the analysis synthesises normative scholarship, institutional guidance documents, and international governance frameworks to identify principles that recur across both domains. Sources were selected for their direct relevance to the governance of high-impact environmental technologies and for their substantive treatment of uncertainty, distributional effects, public participation, and institutional accountability. The analysis draws particularly on the Oxford Principles, the Tollgate Principles (Gardiner & Fragnière, 2018), UNESCO's *Ethics of Climate Engineering* (UNESCO COMEST, 2023), the American Geophysical Union's *Ethical Framework for Climate Intervention Research* (Williams et al., 2026), and scholarship concerning trustworthy, sustainable, and accountable environmental AI.

The comparative analysis proceeds by examining each framework according to four analytical questions. First, how does the framework address scientific uncertainty and the possibility of significant or irreversible harm? Second, how does it address the distribution of risks, benefits, and decision-making power across populations and generations? Third, what forms of participation or consent does it require from affected communities and other stakeholders? Fourth, what mechanisms of transparency, oversight, monitoring, and institutional responsibility does it establish? These dimensions were selected because they recur across the governance literature examined in this study and because each addresses a distinct component of legitimate decision-making under conditions of environmental and technological uncertainty.

The analysis then compares the relative emphasis placed on these dimensions across climate-engineering and environmental-AI governance. Particular attention is given to areas of convergence and to instances in which one literature supplies governance considerations that remain comparatively underdeveloped in the other. The resulting synthesis suggests precaution, justice, participation, and accountability as interconnected dimensions of governance rather than isolated ethical principles.

On this basis, the article develops the Integrated Precautionary Governance Model (IPGM) as a theoretical framework for evaluating high-impact environmental technologies. The IPGM is therefore not presented as an empirically validated predictive model. Rather, it is a conceptual synthesis derived from the comparative analysis of existing governance scholarship and institutional frameworks. Its purpose is to provide an integrated analytical architecture that can subsequently be operationalised and tested across different environmental technologies and institutional contexts.

Results

Evolution of Ethical Governance Frameworks

The Oxford Principles, formulated in 2009, represented the first systematic attempt to articulate ethical guidelines for geoengineering. Their five core commitments — governing geoengineering as a public good, ensuring public participation, requiring public disclosure of research, mandating independent impact assessment, and establishing governance before deployment — were designed as a flexible framework for both research and eventual deployment decisions. Their political context, however, shaped their limitations: as Kruger (2018) has noted, they were

drafted during a period of significant sensitivity surrounding the 2009 Copenhagen climate negotiations, which partly accounts for their procedural rather than substantive character.

Gardiner and Fragnière (2018) have argued most systematically that the Oxford Principles remain predominantly procedural, failing to engage the deeper ethical questions of justice, respect, and democratic legitimacy that geoengineering raises. Their alternative, the Tollgate Principles, insists that proposals must clear substantive ethical thresholds — not merely satisfy transparency and consultation requirements — before proceeding. Morrow et al. (2018) have examined the practical challenges of operationalising such requirements within existing institutional structures, acknowledging their ethical rigor while cautioning about implementation gaps.

One of the most significant recent developments in climate-engineering governance has been the American Geophysical Union's Ethical Framework Principles for Climate Intervention Research, issued in 2024. The framework establishes five principles intended to guide climate-intervention research and emphasizes transparency, inclusivity, justice, and public engagement (Williams et al., 2024; Williams, Shimamoto, & Graumlich, 2024). Subsequent work by Williams et al. (2026) has further discussed the framework's foundations, principles, and ongoing global dissemination. UNESCO's COMEST report, distributed to all 194 Member States, reinforces these themes, identifying knowledge gaps, risks of geopolitical weaponisation, and the danger that climate engineering may divert political will from emissions reduction and adaptation (UNESCO COMEST, 2023). The report explicitly recommends legal obligations to prevent harm, transboundary impact assessment, and the full participation of frontline communities most likely to bear the consequences of both climate change and geoengineering interventions.

Perhaps no concept is more central to the ethics of climate engineering than moral hazard: the concern that the prospect of a planetary thermostat will reduce political and social pressure for emissions reductions. Tsipiras and Grant (2022) identify at least eight distinct framings of this concern in the geoengineering literature, with three predominating: an "Insurance" frame treating geoengineering as a safety net that reduces mitigation urgency; an "Unwilling" frame in which key actors resist mitigation in anticipation of a technical fix; and an "Avoid" frame suggesting geoengineering will be deployed deliberately to circumvent structural economic change.

Smith (2022) situates moral hazard within intergenerational ethics, arguing that geoengineering risks compounding the asymmetry between those who cause climate harm and those who bear it, by enabling present actors to defer decarbonisation while loading future generations with ongoing obligations to maintain planetary-scale interventions indefinitely.

Romero Torres (2021) extends the moral hazard critique by foregrounding its distributional dimensions: geoengineering research is driven overwhelmingly by wealthy-nation institutions, yet its risks, including monsoon disruptions and the potentially catastrophic termination shock of any sudden operational cessation, fall disproportionately on the Global South. Taebi et al. (2023) situate this asymmetry within broader environmental justice frameworks, arguing that the genuine epistemic uncertainties of climate engineering at planetary scales demand a governance architecture that places the burden of proof firmly on deployment proponents, while maintaining mechanisms for reversibility and ongoing independent assessment.

Bridge (2025) identifies climate engineering as among the most ethically complex emerging policy domains, one whose risks cannot be managed through procedural governance alone.

Morrow et al. (2018) established that the Principle of Respect requires global public consent to be secured through governmental representatives before any research with potentially

transboundary environmental effects proceeds, constituting a substantive ethical requirement grounded in the recognition that unilateral action by any subset of states or scientific communities is inherently illegitimate.

The debate over consent in climate engineering governance has crystallised around two positions. The first treats consent as a categorical prerequisite: no actor may deploy or conduct large-scale outdoor experiments without the prior, free, and informed consent of those affected. Pacaol and Tumulak (2026) develop an alternative through the concept of "earned consent," arguing that high-emission states' refusal to consent to interventions aimed at protecting vulnerable populations cannot be treated as morally decisive. Their argument does not endorse unilateral deployment but challenges the philosophical foundations of consent absolutism in contexts of severe and ongoing moral failure, signaling a maturation of the debate toward its harder cases.

Kwakye (2026), drawing on 150 peer-reviewed sources, identifies five ethical domains implicated in climate intervention governance: conceptual foundations, global climate justice, intergenerational and intragenerational ethics, ecological ethics, and ethical governance. He concludes that embedding ethics in climate policy requires equitable finance, inclusive participation, and intergenerational accountability, conditions necessary for both legitimacy and effectiveness. UNESCO COMEST (2023) operationalizes these principles by calling for the full participation of frontline communities, recognising that meaningful inclusion requires actively building institutional capacity rather than extending formal invitations to communities that lack the resources to engage substantively.

Callies (2019) provides a philosophically rigorous account of what distributive equity requires in a world already characterised by profound injustice in both the causes and consequences of

climate change. Solar radiation management governance, he argues, must include compensatory mechanisms for those bearing disproportionate risks, meaningful veto rights for the most vulnerable communities, and robust monitoring of distributional impacts. Landes (2024) extends this by distinguishing feasibility, permissibility, and preferability as distinct evaluative dimensions: most governance frameworks address the first two but sidestep the deeper normative question of whether climate engineering is genuinely preferable to its alternatives, a question that requires explicit commitments about the relative value of different futures and cannot be resolved by procedural means.

Comparative analysis of contemporary governance frameworks indicates a clear evolution in the ethical governance of emerging environmental technologies. Early governance approaches, exemplified by the Oxford Principles, primarily emphasized transparency, public participation, independent assessment, and regulatory oversight. Subsequent frameworks—including the Tollgate Principles, the American Geophysical Union's Ethical Framework for Climate Intervention Research, and UNESCO's *Ethics of Climate Engineering*—expanded this foundation by incorporating distributive justice, intergenerational responsibility, meaningful stakeholder engagement, and institutional accountability. Collectively, these developments demonstrate a transition from governance centred primarily on procedural safeguards toward governance grounded in substantive ethical legitimacy.

The rapid expansion of AI into environmental governance has raised three interconnected ethical concerns: algorithmic biases embedded in environmental data and modelling, the material environmental costs of AI infrastructure, and diffuse or opaque accountability for AI-mediated regulatory decisions. McGovern et al. (2022) demonstrate that environmental data is not inherently objective: AI systems trained on historical records systematically reproduce the

underrepresentation of marginalised communities and geographies, with serious consequences for the distribution of climate risk information and adaptation resources. Their subsequent 2024 work develops a detailed taxonomy of bias types spanning training data, model architecture, evaluation methodology, and deployment context, providing a practical framework for identifying and mitigating bias throughout the AI development lifecycle (McGovern et al., 2024).

Debnath et al. (2023) situate algorithmic bias within the dynamics of planetary-scale climate action, arguing that poorly designed AI systems could undermine the political processes critical to effective mitigation and adaptation. Their proposed response emphasises human-in-the-loop architectures that integrate human judgment and diverse knowledge systems into AI-mediated environmental governance. Vega (2026) identifies a structural contradiction at the heart of AI-driven environmental governance: the same systems promoted as climate solutions may, through their energy demands, resource consumption, and concentrated ownership structures, exacerbate the inequalities and ecological pressures they purport to address. Assessing AI's ethical adequacy for environmental ends, therefore, requires attention to the broader political economy of AI development, not merely the performance of individual applications in isolation.

Convergence of Climate Engineering and Environmental AI Governance

The UN Sustainable Development Goals have emerged as an indispensable organising framework for evaluating both the potential contributions and ethical risks of AI and climate engineering. Vinuesa et al. (2020) establish that AI can support 128 of the 169 SDG targets while potentially inhibiting 58, underscoring that outcomes depend entirely on how AI is designed, deployed, and governed. Sætra (2021) deepens this by insisting that AI must be understood as a sociotechnical system whose SDG impacts include not only direct benefits such as improved

climate modelling and energy optimisation, but also second-order effects, including increased inequality, corporate power concentration, and digital divides that limit Global South access to AI's advantages. Sharma et al. (2026) conclude that realising AI's SDG potential requires a paradigm shift from technosolutionism to justice-centred design, grounded in international cooperation, open-source infrastructure, and inclusive participation. Zaidan et al. (2026) advance this agenda with a proposal for a new SDG 18 on Responsible and Inclusive AI, integrating principles-based, rights-based, risk-based, and ethics-based governance into a hybrid architecture with six measurable global targets.

The emergence of mandatory AI ethics audits represents one of the most consequential recent developments in environmental AI governance, marking a shift from voluntary self-assessment to enforceable third-party evaluation. Ali et al.(2026) provide the most comprehensive framework for this transition, defining algorithmic sustainability through three simultaneously necessary conditions: ecological effectiveness assessed across the full AI lifecycle, institutional accountability anchored in enforceable oversight, and ethical legitimacy grounded in justice and meaningful contestation rights. Züger et al. (2026) complement this with an accessible, interview-based audit methodology designed to produce outputs that civil society can evaluate directly, treating transparency about AI's actual social and environmental impacts as a normative requirement rather than a voluntary disclosure. Zeleti (2026) identifies a critical gap in existing AI trust mechanisms: current certification schemes lack measurable, publicly visible environmental criteria, leaving AI assurance and environmental governance poorly integrated. Mesicek's (2026) SCOPE framework, which assesses AI projects across dimensions of sufficiency, carbon footprint, outcomes, power, and endurance, addresses this gap by requiring that AI deployment be demonstrated as genuinely warranted before proceeding, a direct analogue

of the precautionary logic embedded in the Oxford Principles for climate engineering governance.

Toward an Integrated Precautionary Governance Model

The comparative analysis suggests four governance dimensions that recur across the climate-engineering and environmental-AI literatures: precaution, justice, participation, and accountability. Although these dimensions emerge from different intellectual and institutional traditions, their underlying function is similar. Each addresses a characteristic problem associated with high-impact environmental technologies: uncertainty about consequences, unequal distribution of risks and benefits, exclusion of affected populations from decision-making, and the difficulty of assigning responsibility when technological systems produce harmful outcomes.

The comparative literature suggests that climate engineering and environmental AI share several governance challenges that extend beyond the technical characteristics of either field individually. Historically, environmental governance has often relied on what may be termed a regulatory model, in which oversight mechanisms are primarily reactive and designed to address harms after they occur. More recent approaches, exemplified by the Oxford Principles, shifted attention toward procedural governance, emphasising transparency, public participation, disclosure, and independent assessment. While these innovations improved accountability, critics have increasingly argued that procedural safeguards alone are insufficient when technologies possess global, uncertain, and potentially irreversible consequences (Gardiner & Fragnière, 2018; Morrow et al., 2018).

The first dimension, precaution, responds to uncertainty and potential irreversibility. In climate-engineering governance, precaution is reflected in requirements for prior assessment,

controlled research, independent oversight, and caution concerning deployment. The Oxford Principles and subsequent frameworks emphasise the importance of governance before deployment, while UNESCO and the AGU framework extend these concerns to transboundary effects, independent oversight, and the integration of affected communities into research design (UNESCO COMEST, 2023; Williams et al., 2026). In environmental AI, an analogous concern arises when systems are deployed despite uncertainty regarding model performance, environmental consequences, or downstream social effects. In both contexts, precaution shifts governance from a reactive model—responding to harm after it occurs—to an anticipatory model in which uncertainty itself becomes a consideration in decision-making.

The second dimension, justice, addresses the distribution of risks, benefits, and decision-making power. Climate-engineering scholarship has repeatedly emphasised that those most vulnerable to climate change may have comparatively little influence over the development of climate interventions, while environmental-AI research has demonstrated how biased data, unequal technological capacity, and concentrated ownership can reproduce existing environmental and social inequalities (Callies, 2019; Romero Torres, 2021; McGovern et al., 2022; Vega, 2026). Justice therefore requires governance institutions to consider not merely whether a technology produces aggregate benefits, but who receives those benefits, who bears its risks, and who possesses meaningful influence over decisions concerning its development and deployment.

The third dimension, participation, addresses the legitimacy of decisions affecting populations that may have limited influence over technological development. Climate-engineering governance has increasingly emphasised global consent and the involvement of frontline

communities, while environmental-AI governance has stressed stakeholder participation, contestability, and the incorporation of diverse forms of knowledge (Morrow et al., 2018; UNESCO COMEST, 2023). In both cases, participation must extend beyond symbolic consultation. Meaningful participation requires institutional mechanisms through which affected populations can influence decisions, challenge assumptions, and contribute to the design and evaluation of technological interventions.

The fourth dimension, accountability, addresses the institutional capacity to monitor consequences, assign responsibility, and correct harmful outcomes. In climate engineering, this includes independent assessment, transparency, monitoring, and governance structures capable of responding to transboundary effects. In environmental AI, comparable mechanisms include documentation, ethics audits, independent evaluation, transparency regarding system impacts, and avenues for public redress (Ali et al., 2026; Züger et al., 2026; Zeleti, 2026). Accountability consequently provides the institutional mechanism through which the other three dimensions can be maintained over time.

Taken together, these four dimensions suggest that the governance of high-impact environmental technologies cannot be reduced to procedural compliance or technical risk management alone. Precaution without justice may protect against generalized harm while leaving vulnerable populations disproportionately exposed; justice without participation may identify unequal impacts without giving affected communities meaningful influence over decisions; participation without accountability may create opportunities for consultation without requiring institutions to respond to the concerns expressed; and accountability without precaution may establish

mechanisms for responding to harm without preventing foreseeable risks. The IPGM therefore treats precaution, justice, participation, and accountability as mutually reinforcing conditions of legitimate and sustainable technological governance. Its contribution lies not in proposing four novel ethical principles, but in integrating principles that have often been addressed separately across climate-engineering and environmental-AI governance into a common analytical architecture.

Application of the IPGM

The practical value of the IPGM becomes evident when applied to emerging proposals for mandatory ethics audits of high-risk environmental AI systems. Increasingly, artificial intelligence is being deployed to support decisions involving wildfire prediction, water allocation, biodiversity conservation, climate adaptation, and ecosystem restoration—contexts in which algorithmic recommendations may influence the distribution of environmental risks and resources across entire communities. Existing governance approaches often evaluate such systems primarily through measures of technical performance, regulatory compliance, or post-deployment monitoring. An integrated precautionary approach would require a broader assessment. Precaution demands independent evaluation of scientific uncertainty and the potential consequences of algorithmic error before deployment. Justice requires consideration of whether system outputs disproportionately affect vulnerable populations or reinforce existing environmental inequalities. Participation requires meaningful consultation with affected stakeholders throughout the design, implementation, and review of AI systems, while accountability requires transparent documentation, independent ethics audits, mechanisms for public oversight, and clear avenues for institutional redress. Viewed through this framework, these audits go beyond simple procedural compliance exercises; they serve as an institutional

mechanism for integrating scientific reliability with democratic legitimacy and distributive justice in the governance of environmental technologies.

Discussion

Governance Implications

The ethical frameworks shaping environmental policy are more institutionally grounded and more attentive to questions of justice than their predecessors. The Oxford Principles have evolved into a richer governance architecture, supplemented by the AGU's Ethical Framework, UNESCO's COMEST recommendations, and a growing scholarly consensus that demands substantive, rather than merely procedural, ethical standards. The moral hazards of climate engineering and AI-driven environmental governance are better understood than at any previous moment. The pursuit of global consent for high-stakes planetary interventions has increasingly emerged as a central governance principle, and the shift toward mandatory bias mitigation and ethics auditing in environmental AI represents a genuine paradigm shift in how technological accountability is conceptualised.

Implications for Sustainability Governance

The IPGM has implications that extend beyond the governance of climate engineering and environmental AI. At its core, the framework proposes that sustainability governance should evaluate technological interventions not only according to whether they produce desirable environmental outcomes, but also according to whether the institutions governing them are capable of addressing uncertainty, distributing risks and benefits equitably, incorporating affected populations, and responding to harmful consequences. This shifts the evaluation of emerging environmental technologies from a narrow question of technical effectiveness toward a broader

assessment of whether technological transformation is institutionally and socially sustainable.

First, the IPGM places precaution at the centre of sustainability governance. Conventional environmental regulation frequently responds to demonstrated harms, whereas emerging technologies may create consequences that are difficult to predict or reverse. A sustainability-oriented governance system should therefore require meaningful assessment before deployment and treat significant uncertainty as a reason for additional scrutiny rather than as a justification for assuming that technological benefits will outweigh potential harms.

Second, the framework establishes justice as an essential component of sustainability rather than a secondary consideration. The distribution of technological risks and benefits is inseparable from the sustainability of environmental interventions. Climate engineering may affect populations that have contributed relatively little to climate change, while environmental AI may distribute environmental resources or regulatory burdens through systems trained on historically unequal data. Sustainable technological transformation therefore requires attention to distributional, procedural, and intergenerational equity.

Third, participation provides a mechanism for connecting sustainability governance with democratic legitimacy. Environmental technologies increasingly affect communities that may have limited influence over their design or deployment. Meaningful participation requires more than information disclosure or consultation after decisions have been made. It requires institutional arrangements through which affected communities can contribute to governance, contest proposed interventions, and influence decisions concerning acceptable levels of risk.

Finally, accountability determines whether these principles can be maintained after technological systems enter practice. Impact assessments, independent ethics audits, transparent documentation, monitoring, and mechanisms for institutional redress provide ways of ensuring that governance does not end when a technology is approved. Sustainability governance must therefore possess the capacity to learn from consequences, identify emerging harms, and modify or discontinue interventions when necessary.

Viewed collectively, these implications position the IPGM as a framework for sustainable technological transformation. Its central proposition is that environmental technologies should not be considered sustainable solely because they address an environmental problem. Their sustainability also depends on the institutions through which they are developed, evaluated, deployed, monitored, and governed. This perspective connects technological innovation with the broader sustainability objectives of equity, participation, institutional resilience, and long-term environmental responsibility.

Remaining Challenges

Yet these advances do not dissolve the fundamental tensions in environmental technology governance. The concentration of technological capability in wealthy nations continues to generate asymmetric risks for the Global South. The pace of technological development continues to outstrip the capacity of governance frameworks to anticipate, evaluate, and constrain it. And the question of whether technical fixes supplement or substitute for the structural transformation of energy and economic systems remains as politically contested as

ever. Recent ethical frameworks offer substantially stronger foundations for governing emerging environmental technologies than many earlier approaches.

Limitations

This study employs conceptual comparative analysis rather than empirical investigation. Consequently, the IPGM should be understood as a theoretical governance framework derived from synthesis of existing scholarship rather than an empirically validated governance model. Its four dimensions—precaution, justice, participation, and accountability—represent a conceptual synthesis of the frameworks examined rather than a claim that these dimensions exhaust all possible requirements of legitimate environmental governance.

The analysis is also necessarily shaped by the selection of governance frameworks and scholarship examined. The sources considered were selected for their direct relevance to climate engineering, environmental AI, sustainability, and emerging technology governance, but the analysis does not constitute a systematic review of all literature addressing these fields. Different source selections could therefore identify additional principles or place different emphasis on the four dimensions proposed here. Future research should test the robustness of the IPGM through broader systematic review and comparative analysis across additional technologies and institutional settings.

A further limitation concerns operationalisation. The IPGM identifies four dimensions of legitimate governance but does not establish quantitative thresholds for determining when a particular technology satisfies them. Questions concerning how much participation is sufficient, what degree of uncertainty should trigger precautionary restrictions, how distributive impacts

should be measured, and what institutional mechanisms constitute adequate accountability remain unresolved. These questions are important because a framework that identifies desirable governance principles without specifying how they can be evaluated risks remaining primarily normative.

The IPGM should therefore be regarded as a framework for organising subsequent empirical and policy research rather than as a completed governance instrument. Future research should develop indicators for each dimension, examine how the dimensions interact in practice, and evaluate whether their combined application improves the legitimacy and sustainability of decisions concerning high-impact environmental technologies.

Conclusion

Climate engineering and environmental AI represent increasingly consequential forms of environmental intervention. Their significance extends beyond their technical capabilities because both can redistribute environmental risks and benefits, influence political decision-making, and produce consequences that cross national and generational boundaries. Their governance therefore requires more than technical assessment or conventional regulatory compliance.

This article has argued that contemporary governance frameworks across climate engineering and environmental AI exhibit substantial overlap around four mutually reinforcing requirements: precaution, justice, participation, and accountability. The Integrated Precautionary Governance Model (IPGM) brings these dimensions together as a common framework for evaluating the legitimacy and sustainability of high-impact environmental technologies. Its contribution is not

the invention of four new ethical principles, but the integration of principles that have frequently been addressed separately across different governance literatures.

The IPGM also advances a broader conception of sustainability governance. A technology cannot be considered sustainable solely because it promises environmental benefits. Its sustainability depends in part upon whether the institutions governing it can anticipate uncertainty, address unequal distributions of risk and benefit, include affected populations in meaningful decision-making, and remain accountable for consequences after deployment. Sustainable technological transformation therefore requires governance institutions capable of integrating environmental objectives with justice, democratic participation, and long-term institutional responsibility.

The framework remains conceptual and requires empirical development. Future research should operationalise its four dimensions, develop indicators for assessing their implementation, and test the IPGM across additional environmental technologies and institutional settings, including large-scale carbon removal, biodiversity engineering, autonomous environmental monitoring, and other emerging interventions. Comparative research could also examine whether governance systems that integrate precaution, justice, participation, and accountability produce more legitimate and resilient sustainability outcomes than approaches centred primarily on technical performance or procedural compliance.

The broader implication is that the central governance question for emerging environmental technologies is not whether societies must choose between technological innovation and precaution. Rather, it is whether institutions can develop the capacity to make technological

innovation compatible with environmental sustainability, distributive justice, democratic legitimacy, and long-term accountability.

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Declaration of AI Usage

Grammarly was used solely for basic spelling, grammar, punctuation, and proofreading assistance. No generative AI tools were used to develop, generate, or substantively revise the article's arguments, analysis, framework, or conclusions.

Data Availability Statement

This study did not generate or analyse a new dataset. The analysis is based exclusively on publicly available scholarly literature and institutional governance documents cited in the manuscript.

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